ALICE L WALTON SCHOOL OF MEDICINE

# Nonopioid Analgesics–NSAIDS

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Adapted from: Nonopioid Analgesics

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0:00 / 29:58 1x

Learning Objectives (6) Versions

*After completing this brick, you will be able to:*

1. Explain the mechanism of action of NSAIDs including the specific pharmacological targets and end products within the arachidonic acid pathway.
2. Classify NSAIDs based on their COX enzyme selectivity and type of inhibition and discuss the clinical implications.
3. Compare an… based on their therapeutic dose, therapeutic effects, and associated side effects.
4. Explain the potential side effects of NSAIDs, focusing on their impact on the gastrointestinal, cardiovascular, and renal systems, and highlight other notable adverse effects.
5. Identify common drug interactions involving NSAIDs, elucidating the underlying pharmacological mechanisms and the potential clinical consequences for co-administration with other medications.
6. Apply the knowledge of NSAIDs pharmacology to provide rationale for evidence-based recommendations in various clinical scenarios.

CASE CONNECTION

GL is 35-year-old man at the emergency department for treatment of acute upper GI bleeding. He reports, "I was doing okay until today. I had some stomach pains, but nothing I couldn't deal with. I've been getting over my knee surgery, taking ibuprofen, and it's been working. Then I started vomiting blood. With my leg in a cast, I couldn't get to the bathroom in time to vomit. My living room is a mess." His pulse is 110/min and blood pressure 92/60 mm Hg. He vomits about 100 cc of red blood during your examination. His skin is pale. His abdomen is tender over the epigastric region, with no rebound tenderness.

What is likely to have caused GL's bleeding? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

## Nonsteroidal Anti-inflammatory Drugs

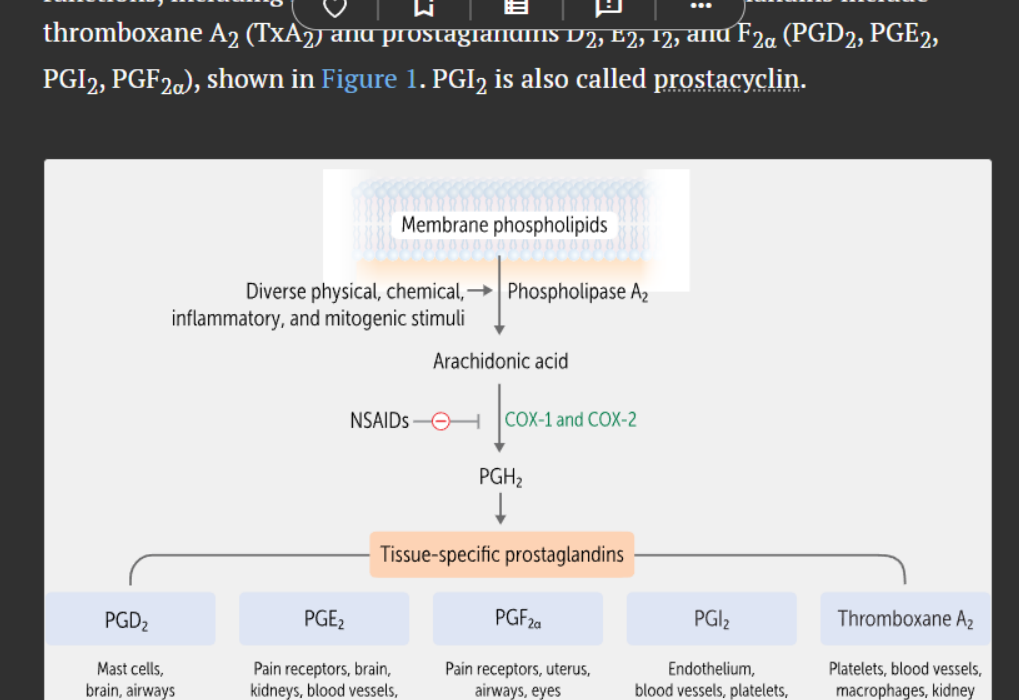
Nonsteroidal anti-inflammatory drugs (NSAIDs) are medications used to treat both mild to moderate pain (analgesic), inflammatory pain conditions, such as rheumatoid arthritis or acute gout (anti-inflammatory) and can reduce fever (antipyretic). As their name implies, they work by reducing inflammation. How? NSAIDs block two key enzymes, cyclooxygenase 1 (COX-1) and cyclooxygenase 2, that promote inflammation. Common NSAIDs include aspirin and ibuprofen.

### The Inflammatory Pathway

In response to tissue damage, pro-inflammatory cells such as neutrophils and macrophages interact with the tissue membranes, causing release of membrane phospholipids, one of which is arachidonic acid. This is an abundant membrane phospholipid and a key substrate in the inflammatory pathway. Once arachidonic acid is released from the membrane, it serves as a substrate for two key enzymes, COX-1 and COX-2:

- COX-1 is referred to as the constitutive form (the form continuously expressed to maintain body homeostasis). It helps maintain many helpful body functions, including dilation of renal arteries and secretion of gastric mucous.
- COX-2 is referred to as the inducible form. This is because it is mostly expressed during inflammation or tissue injury. This can increase pain.

Both COX-1 and COX-2 cleave arachidonic acid into prostaglandin $H_2$, which is further converted into other prostaglandins (PG) that mediate a variety of body functions, including … …glandins include thromboxane $A_2$ ($TxA_2$) and prostaglandins $D_2$, $E_2$, $I_2$, and $F_{2\alpha}$ ($PGD_2$, $PGE_2$, $PGI_2$, $PGF_{2\alpha}$), shown in Figure 1. $PGI_2$ is also called prostacyclin.


Membrane phospholipids
Diverse physical, chemical, inflammatory, and mitogenic stimuli
Phospholipase A2
Arachidonic acid
NSAIDs
COX-1 and COX-2
PGH2
Tissue-specific prostaglandins
PGD2
Mast cells, brain, airways
PGE2
Pain receptors, brain, kidneys, blood vessels, platelets
PGF2α
Pain receptors, uterus, airways, eyes
PGI2
Endothelium, blood vessels, platelets, kidney, brain
Thromboxane A2
Platelets, blood vessels, macrophages, kidney

QUIZ

Figure 1

Credit: ©ScholarRx

COX-1 and COX-2 catalyze oxidation of which phospholipid to form prostaglandin $H_2$?

Let's look at some key functions of these prostaglandins, which will become important as we think about the effects of drugs that block them:

- Inflammation: All of the PG in Figure 1 increase the inflammatory response (eg, pain, heat, swelling, redness). This mostly occurs using the COX-2 pathway, because that is the one induced during inflammation and tissue injury.
- Pain: $PGF_{2\alpha}$ and $PGE_2$ both stimulate pain receptors.
- Platelet function: $TxA_2$ causes platelets to aggregate and increases clot formation, whereas $PGI_2$ (prostacyclin) inhibits it.
- Gastric acid secretion: $PGE_2$ reduces gastric acid secretion and causes the production of protective gastric mucous. Both actions protect the stomach from acid injury.
- Blood pressure and flow: $PGE_2$ and $PGI_2$ both dilate blood vessels, whereas $TxA_2$ constricts them.

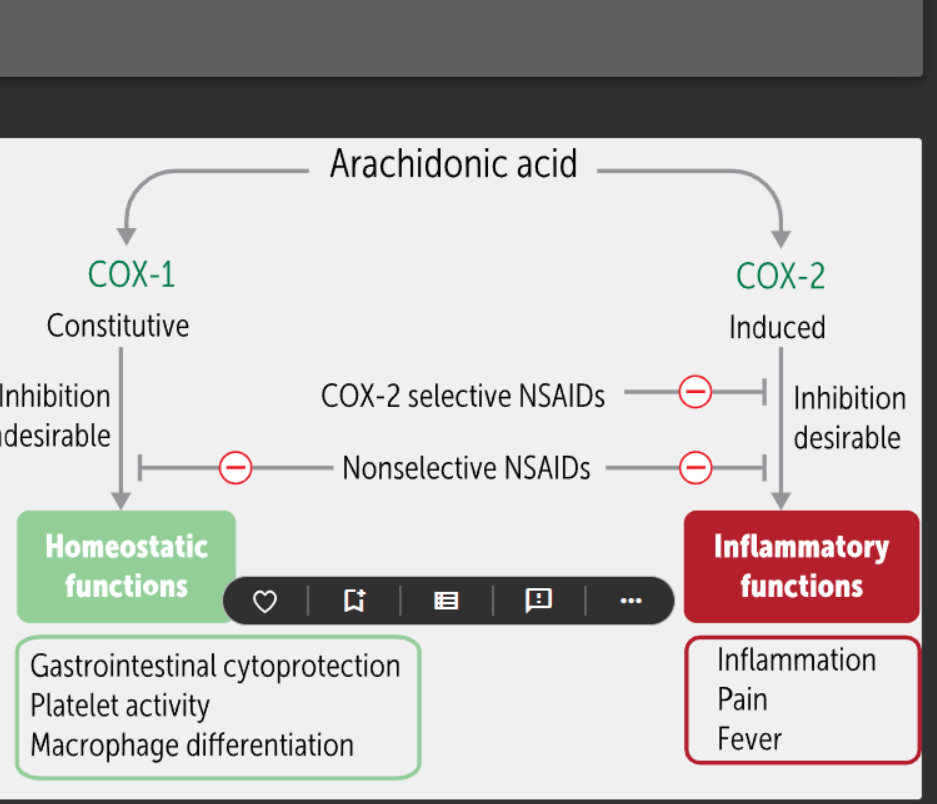
Which prostaglandin inhibits platelet aggregation?

## HOW DO NSAIDS WORK? (LO#1)

NSAIDs block either both COX-1 and COX-2 (nonspecific inhibitors) or just COX-2, inhibiting formation of prostaglandins (Figure 2). COX-2 inhibition reduces inflammation and pain, so these drugs are used for many inflammatory conditions like arthritis, as well as general pain. It is important to note that most NSAIDs are competitive, reversible inhibitors of the COXs. However, Aspirin inhibits the enzyme irreversibly. There are two types of NSAIDs, nonselective and selective:

- **Nonselective NSAIDs** are most common and inhibit both COX-1 and COX-2.
- **Selective NSAIDs** have a higher affinity towards COX-2. These drugs were developed because it was hoped that selective blockage of COX-2 would lead to the therapeutic anti-inflammatory action of NSAIDs without the undesirable adverse effects of nonselective NSAIDs due to their inhibiting some of the normal body functions discussed above (like renal vasodilation and gastric prote… …partially successful, and selective COX-2 inhibitors have their own problems (see below).

What is the primary difference between nonselective and selective NSAIDs?



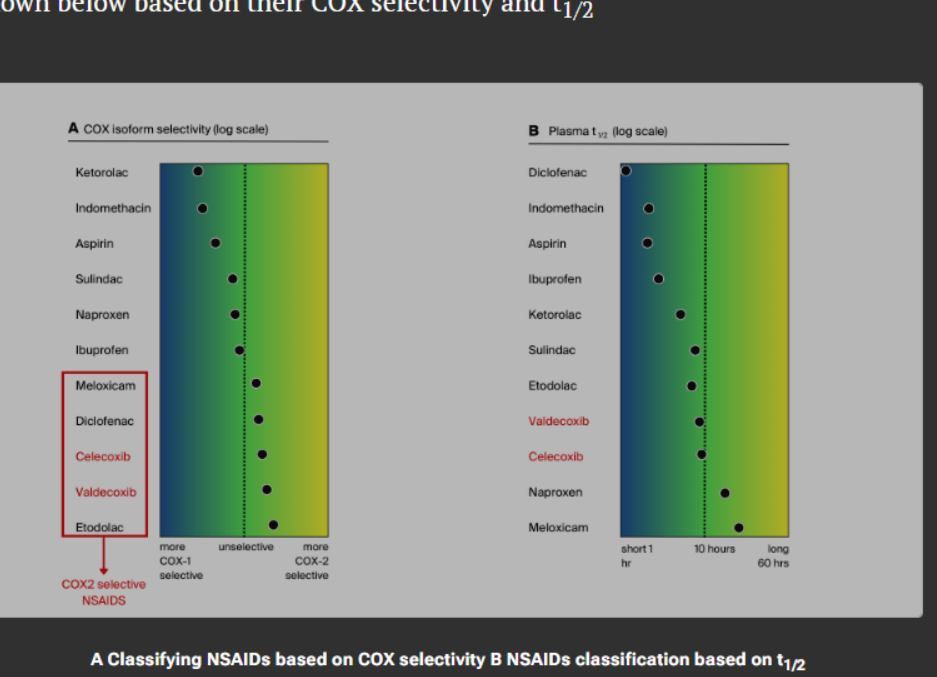
QUIZ

Figure 2

What is the difference between COX-1 and COX-2?

## HOW DO WE CLASSIFY NSAIDS? (LOs#2,3)

There are few ways to classify NSAIDs. The most common classification is by COX inhibition selec… …COX inhibitors (inhibit both COX-1 and -2). With agents specifically designed for COX-2 inhibition, COX-2-selective NSAIDs (ending with—coxibs) were easy to categorize. However, it was soon discovered that some older NSAIDs (such as Diclofenac, Meloxicam) were also COX-2-selective. Commonly used NSAIDs are shown below based on their COX selectivity and $t_{1/2}$



A Classifying NSAIDs based on COX selectivity B NSAIDs classification based on $t_{1/2}$

Aspirin (acetylsalicylic acid), an NSAID has unique MOA and clearly distinguished by its chemical structure. NSAIDs are also classified as:

- **Aspirin or salicylates** (acetylsalicylic acid) irreversibly inhibits COX-1 and COX-2. It is the most potent inhibitor of platelet aggregation, so it is used in clot prevention… …aving a myocardial infarction, or those with a history of stroke and atrial fibrillation, to prevent further clotting. Aspirin is discussed in further detail in a separate brick.
- **Nonsalicylate NSAIDs** include ibuprofen, naproxen, indomethacin, and many others. These NSAIDs are chemically derived from other carboxylic acids. They are commonly used as inexpensive, over-the-counter pain relievers and anti-inflammatory agents but are not used as platelet inhibitors like aspirin. Indomethacin can also be used in premature neonates to close a patent ductus arteriosus (PDA).

Which NSAID is also used as an antiplatelet medication, to prevent a stroke?

## WHAT ARE THE INDICATIONS FOR NSAID USE? (LOs#3,6)

- ***Inflammation (ar…*** …ure 1 increase the inflammatory respo… …ness). This mostly occurs using the COX-2 pathway, because that is the one induced during inflammation and tissue injury. The downstream inhibition of NF-κB also decreases production of pro-inflammatory factors. Thus, NSAIDs are used as anti-inflammatory agents.
- ***Fever (anti-pyretic):*** COX inhibition by NSAIDs interferes with the PGE2 mediated reset thermoregulation in the hypothalamus. Disruption of this process helps to prevent increase in core body temperature.
- ***Pain (analgesic):*** NSAIDs can be used as an analgesic, with the inhibition of PGF2α and PGE2. (both stimulate pain receptors).
- ***Platelet function (anti-thrombotic):*** TxA2 causes platelets to aggregate and increases clot formation, whereas PGI2 (prostacyclin) inhibits it. Thus, they can used as anti-thrombotics.
- ***Fetal Circulatory System:*** PGs help in maintaining the patency of ductus arteriosus (PDA). Indomethacin and Ibuprofen (IV) has been used in neonates to close an inappropriately patent ductus.

ACTIVE LEARNING QUESTION

**Match the regulatory factor to the effect produced by NSAID-mediated COX inhibition**

| | |
|---|---|
| Anti-pyretic | ✓ PGE2 |
| Anti-inflammatory | ✓ NF-kB |
| Anti-Thrombotic | ✓ TxA2 |

**Explanation**

NSAIDs inhibit COX and downstream production of PGs. Specific PGs carry out various functions in the body. Inhibition of these PG-mediated effects yields therapeutic benefits. Anti-pyretic action is mediated by the inhibition of PGE2, anti-inflammatory by NF-KB decrease, and anti-thrombotic by TXA2 inhibition.

CLINICAL CORRELATION

The ductus arteriosus (DA) is a fetal vascular connection between the main pulmonary artery to the aorta, redirecting blood away from the lungs. After birth, the DA actively constricts and eventually closes. If the DA does not fully close after birth, it results in a condition known as patent ductus arteriosus (PDA). The fetal ductus arteriosus (DA) remains open due to low arterial oxygen levels and the presence of prostaglandin E2 (PGE2), which is partially produced by the placenta The reported prevalence of an isolated PDA among term infants ranges from 3 to 8 per 10,000 live births. Acetaminophen along with NSAIDs is also used in the treatment of DA. IV Ibuprofen us is common. Indomethacin use has declined since early 2000s given the safety profile of Ibuprofen

## WHAT ARE THE … …TS ASSOCIATED WITH NSAID USE? (LOs#4,6)

Because NSAIDs are commonly used, it is important to understand their adverse effects and contraindications. As we've already learned, NSAIDs block the COX enzyme cascade, which produces PG that act within the gastrointestinal (GI) tract, kidneys, and heart. So, blocking the formation of these PG, especially by blocking COX-1, can cause adverse effects in these organ systems.

***Gastrointestinal:*** Nonselective NSAIDs block the protective effects of COX-1. Normally, COX-1 produces PGI2 and PGE2 that inhibit gastric acid secretion and promote gastric mucus secretion. This mucus plays a key role in protecting the stomach's lining from gastric acid. Therefore, without this layer of protection, the stomach, esophagus, and small bowel are vulnerable to damage from the gastric acid. In some cases, contact with the gastric mucosa also irritates the lining and induces tissue damage. The use of NSAIDs can lead to the development of NSAID-induced peptic ulcers. For patients who have a history of peptic ulcer disease, the risk of gastric ulcer should be reduced. **Selective COX-2 inhibitors** (instead of non-selective) could be an option. If patient has an active peptic ulcer disease, it is best to avoid all NSAIDs.

How do nonsel… …stinal system?

***Cardiovascular:*** The increased cardiovascular risk associated with NSAID use arises from the inhibition of COX-2 and the downstream inhibition of PGI2. The reduced production of PGI2 decreases the inhibition of the prothrombotic effects on platelet activation, which could lead to potential endothelial injury. Even though COX-2 inhibitor such as Celecoxib is approved for use, it should be avoided in patients with cardiovascular disease or multiple cardiovascular risk factors. COX-2 inhibitors decrease prostacyclin, which prevents platelet aggregation and vasoconstriction. This increase in clot formation and vasoconstriction can lead to thrombi, increased risk of hypertension, and myocardial infarction. In fact, in patients with established coronary heart disease, long term use of NSAIDs is not recommended. However, it is important … …te NSAID) has been shown to … …rdial infarction event … **…out GI or CV risk,** Naproxen … …fen, and Diclofenac posed the **…high risk of CV disease;** … …risk, using the lowest … …ended. In patients without a … …mended.

**coronary heart disease**

Narrowing of the coronary arteries, usually as a result of atherosclerosis, which compromises the supply of oxygen-rich blood to the heart leading to chest pain (angina).

Rx Bricks

Vasculature and Lymphatics of the Heart

Search Term

coronary heart disease
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…atients with cardiovascular

In a patient without any recent cardiovascular event, which NSAID would be appropriate option (s)

***Renal:*** Within the renal system, NSAIDs can lead to reduced glomerular filtration rate and cre… …ney, PGE2 and PGI2 … …lies the glomerulus with … …rostaglandins, causing the … This reduces glomerular … …merular filtration rate (GFR). … kidney injury. It's completely … …n, NSAIDs can cause acute … …dney disease. NSAIDs also … …staglandin-induced renal … , NSAIDs are contraindicated in patients who have significant renal impairment.

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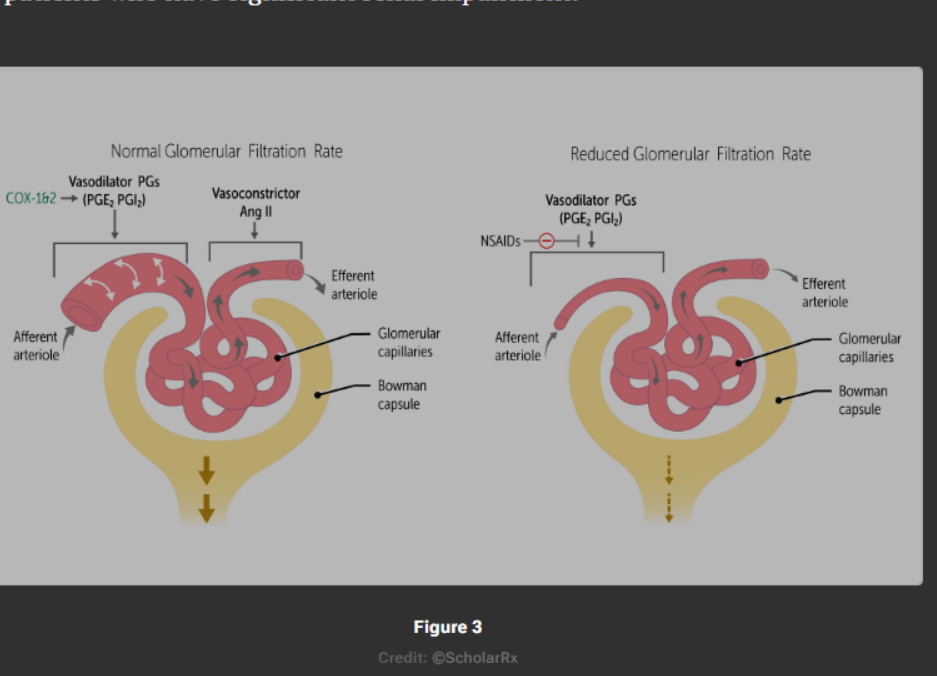

Figure 3

Credit: ©ScholarRx

coronary heart disease
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…Injury (AKI). Renal injury from … …risk decreases with … …mediated-AKI include volume … Kidney Disease, use of … …ge (>65).

***Hypertension:*** NSAIDs may increase blood pressure in patients who are hypertensive, both due to direct effects on the kidney and possibly in reducing the effects of some antihypertensive drugs. Nonaspirin NSAIDs should be used with caution in patients with severe or uncontrolled hypertension.

***Ductus Arteriosus:*** NSAIDs can cause premature closure of ductus arteriosus leading to HF in the baby. NSAIDs are not used in the 3rd trimester of pregnancy (>30 weeks).

Why should NSAIDs be avoided in the 3rd trimester of pregnancy?

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…rse effects. NSAIDs are … …nous (IV), intramuscular (IM), … have systemic absorption. … …c therapy, but if patients are unable to tolerate medications by mouth, IV and IM formulations may be used. However, because of the systemic absorption, these types of NSAIDs have the potential to cause drug interactions and other adverse effects. Topically applied NSAIDs penetrate local tissues in the area of application, which limits systemic absorption. Hence, fewer drug interactions and associated side effects are recorded with topical use. In fact, topical NSAID preparations have shown relative safety and positive benefits. For example, diclofenac gel has been used in the treatment of osteoarthritis. Dosing for NSAIDs for inflammatory conditions with continuous use should start at the lowest effective dose and then increased to the obtain the maximal anti-inflammatory effect.

CLINICAL CORRELATION

NSAIDs are available over the counter and can be found in the medicine cabinets in most households. Providing patient education material is important when prescribing NSAIDs. Resources that provide information at a 5th or 6th grade reading level provide basic information to the patients regarding dosing, use, and adverse effects. Data… …Micromedex provide patient education m…

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## …IONS WITH NSAID USE?

… Risk of *bleeding* and … …ons when thinking about use … …tant in elderly people and in … …s or hypertension.

**Drugs that can increase *bleeding* risk with NSAIDs**

- *Anticoagulants*: Non-salicylate NSAIDs are highly protein-bound and can displace protein-bound drugs such as Warfarin and enhance their effects. Additionally, NSAIDs can also inhibit platelet function. Thus, the use of NSAIDs should be avoided in patients with Warfarin or other anticoagulants to prevent bleeding risk.
- *Glucocorticoids*: The use of glucocorticoids and NSAIDs can increase the risk of GI bleeds.
- *Selective serotonin reuptake inhibitors (SSRIs)*: SSRIs, in combination with NSAIDs, can increase gastroduodenal toxicity.

**Drugs that can increase *nephrotoxicity* risk with NSAIDs**

- *Angiotensin-converting enzyme inhibitors (ACE-I)*: The use of NSAIDs with ACE-I can increase the risk of AKI (described earlier) and hyperkalemia.
- *Others*: All classes of NSAIDs can reduce renal excretion of medications such as lithium (a medication commonly used for bipolar disorder and excreted primarily through the kidneys) and methotrexate (an immunosuppress… …s, psoriasis, and other autoimmune diseases), leading to toxicity.

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…n patients taking ACE-I?

CLINICAL CORRELATION

Most COX-2 inhibitors were withdrawn from the market in the early 2000s because of an increased risk of heart attacks and stroke with prolonged usage.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to GL, what is likely to have caused his bleeding?

As your team of nurs… …blood for tests, you explain the situation … …of your stomach … …ne tests to assess the severity … …need a blood transfusion, and … …have you seen by the … …help stop the bleeding." … …nent room, you overhear GL's … …icine with food and to speak … …n this!"

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## Summary

- Nonsteroidal anti-inflammatory drugs (NSAIDs) are used to treat mild to moderate pain, inflammatory conditions, and fever by blocking enzymes COX-1 and COX-2 that promote inflammation.
- NSAIDs can be classified as nonselective, inhibiting both COX-1 and COX-2, or selective, having a higher affinity towards COX-2.
- Common NSAIDs include aspirin, ibuprofen, and naproxen, each with unique uses and side effects.
- NSAIDs are used for various indications including inflammation, pain, platelet function, and in neonates to close a patent ductus arteriosus.
- Adverse effects of NSAIDs can include gastrointestinal issues, cardiovascular risks, renal impairment, hypertension, and premature closure of ductus arteriosus in babies.
- NSAIDs can interact with other drugs, increasing the risk of bleeding when used with anticoagulants, and causing toxicity when used with drugs like lithium and methotrexate that are excreted through the kidneys.

## Review Questions

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…ID

…AID

D. ● Non-selective COX inhibitor, Nonsalicylate NSAID

**Hide Explanation**

The correct answer is Non-selective COX inhibitor and nonsalicylate NSAID (D). Ibuprofen does not bind selectively to COX-1 (B) or COX-2 (C) As a derivative of salicylic acid, aspirin is the only salicylate NSAID. Hence, selective COX-1 inhibitor, salicylate NSAID (C) is not the correct answer

**2. The chronic use of NSAIDs concomitantly with ACE-I could cause**

A. ○ Gastrointestinal side effects
B. ● Renal side effects
C. ○ Cardiovascular side effects
D. ○ Hepatic side effects

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…se of NSAIDs with ACE-I can … C), or Hepatic (D) have not … …dications.

**3. Which of the following Nonsalicylate NSAID would be best for patients without any GI or CV risk?**

A. ● Naproxen
B. ○ Celecoxib
C. ○ Aspirin
D. ○ Diclofenac

**Hide Explanation**

The correct answer is Naproxen (A). Non-selective COX inhibition have better GI and CV safety. Celecoxib (B), while has lower GI side effects, but increases CV risk. Aspirin (C) is a salicylate and has high GI side effect. Diclofenac (D) has high CV risk.

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**…ndicated in the treatment of**

D. ○ Diclofenac

**Hide Explanation**

The correct answer is Ibuprofen (A). Ibuprofen (IV) and Indomethacin are recommended in the treatment of Ductus arteriosus. Celecoxib is a selective COX-2 inhibitor (B) and is not recommended in the treatment of Ductus arteriosus. Aspirin is a non-selective NSAID, but is not recommended for treatment of Ductus arteriosus (C). Diclofenac is more of a selective COX-2, but it is not recommended for Ductus arteriosus (D).

**5. All nonsalicylate NSAIDs inhibit the COX enzyme**

A. ● competitive and reversibly
B. ○ noncompetitive and reversibly
C. ○ noncomp…

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…ition of COX (A). All Non… …itors. NSAIDs do not bind to … ) and (C) are incorrect. Although NSAIDs bind to COX competitively, they are reversible inhibitors. Hence, answer choice (D) is incorrect. (B).

**How would you rate this Brick?**

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of NSAIDs including the specific pharmacological targets and end products within the arachidonic acid pathway.~~
02 ☑ ~~Classify NSAIDs based on their COX enzyme selectivity and type of inhibiti…~~

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…ectives
…table with the concept
01 ☑ ~~…IDs including the specific pharmacological targets and end products within the arachidonic acid pathway.~~

02 ☑ ~~Classify NSAIDs based on their COX enzyme selectivity and type of inhibition and discuss the clinical implications.~~
03 ☑ ~~Compare and contrast commonly prescribed NSAIDs based on their therapeutic dose, therapeutic effects, and associated side effects.~~
04 ☑ ~~Explain the potential side effects of NSAIDs, focusing on their impact on the gastrointestinal, cardiovascular, and renal systems, and highlight other notable adverse effects.~~
05 ☑ ~~Identify common drug interactions involving NSAIDs, elucidating the underlying pharmacological mechanisms and the potential clinical consequences for co-administration with other medications.~~
06 ☑ ~~Apply the knowledge of NSAIDs pharmacology to provide rationale for evidence-based recommendations in various clinical scenarios.~~

**Objective Confidence: 6/6**